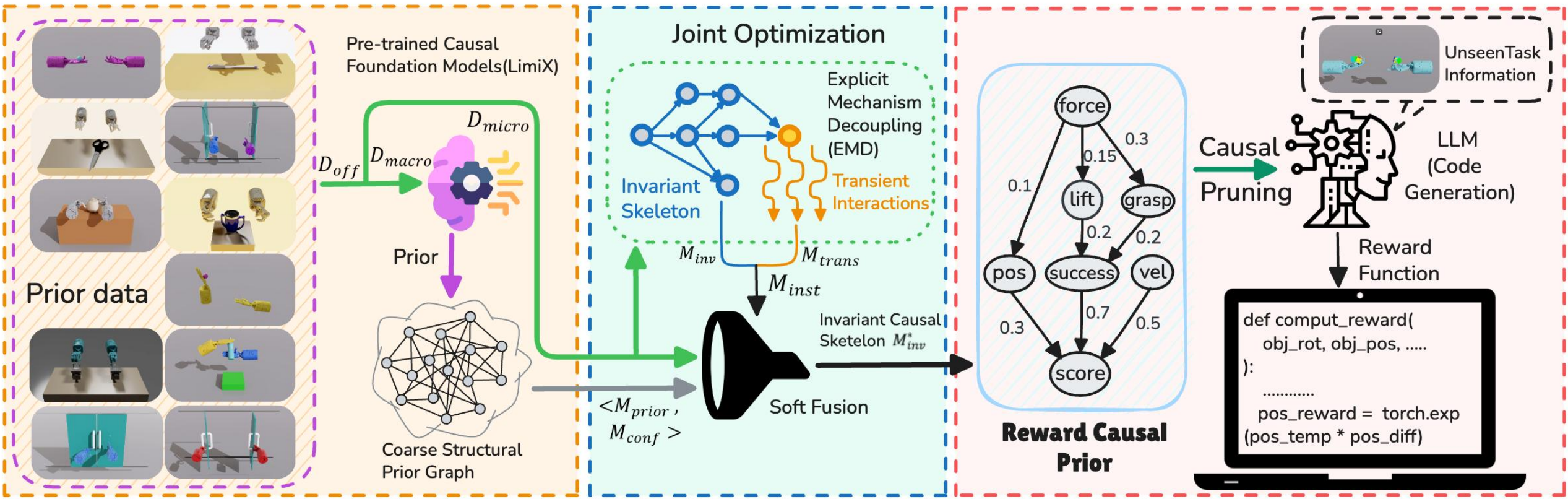

Prior data
Pre-trained Causal Foundation Models(LimiX)
D_{off}
D_{macro}
D_{micro}
Prior
Coarse Structural Prior Graph
Joint Optimization
Explicit Mechanism Decoupling (EMD)
Invariant Skeleton
Transient Interactions
M_{inv}
M_{trans}
M_{inst}
$<M_{prior}, M_{conf}>$
Soft Fusion
Invariant Causal Sketelon M^*_{inv}
force
0.1
0.15
0.3
lift
grasp
0.2
0.2
pos
success
vel
0.3
0.7
0.5
score
Reward Causal Prior
UnseenTask Information
Causal Pruning
LLM (Code Generation)
Reward Function
def comput_reward(
obj_rot, obj_pos, ...
):
.....
pos_reward = torch.exp
(pos_temp * pos_diff)